Gestation

Carrying out the gestation process outside a mother's womb is difficult. Once the embryo implants to the uterus, there occurs the formation of the all important placenta and amniotic sac, filled with amniotic fluid. The placenta is basically a direct connection between the mother and the baby, allowing for exchange of nutrients, hormones and expulsion of waste. The amniotic fluid is like a shock absorbing fluid which also doubles as chemical exchange medium. Simple enough? Not really. There are plenty of uncertainties involved. What is the sequence of these developments? What exactly is transferred, when and how much? The list goes on. Besides, a single mistake in the hormone levels or genetic transfer could have consequences in the ultimate development that are impossible to predict. The current record for the survival of a premature baby is just around 21 weeks into the gestation.
It is with the hope of increasing this survival rate that scientists began to originally think of creating artificial wombs. And, while there are many groups across the world now working in this area of cutting edge biology and medicine, two groups have done seminal research.

First, an artificial womb has to provide a mechanically safe housing for the growing foetus and protect it against infection. Then scientists have to replicate the functions of the placenta, namely to remove toxins and ensure transport of oxygen and hormones. An artificial placenta will also have the capability to regulate blood sugar levels, and do other tasks that the babies newly developed organs might not be initially capable of doing. It's also critical to ensure that the foetus is able to breathe without the support of the mothers lung. This is a major problem in premature babies - lungs too weak to completely support themselves. Any artificial placenta will also have lung enhancement additions or perhaps make use of state of art liquid ventilation technology. Lastly, the functions of the amniotic fluid have to be replicated - which in itself is no mean task. Scientists also remark that the womb will have to mimic the sounds and rhythms of the womb.


A neonatal intensive care unit


Studies of a human foetus by Leonardo Da Vinci

Notable work

Dr. Yoshinori Kuwabara initially lead the charge for development of artificial placenta for a long time now. His team in Japan have been able to sustain growth of goat foetuses in a 'ectogenic chamber'. Then there is Dr. Helen Liu at Cornell university who has devised a sophisticated system for culturing mouse embryos. To sidestep the limitations imposed by the fourteen day rule her group switched from human embryos to experimenting with mouse embryo's. Bit by bit her group has been making additions integral in improving the viability of her artificial wombs and while there has been progress it's a difficult road ahead.

Part of the difficulty lies also in surrounding the embryo with the right fluid, containing the correct percentages of water, hormones, proteins and other essential chemicals. Then these percentages have to be modulated according to how far the embryo is in its development. Also, while success with mouse embryo's is a great step, scaling up to human embryo's will need a paradigm shift in technology.

It goes without saying that this line of research can be vastly beneficial to mankind. If successful, maybe artificial wombs might even become safer than natural births in some cases. The technology will also allow couples that cannot have children by natural means to have babies – free of surrogacy.
But as radical this technology is by its nature, it also raises some fundamental questions. Is the mere fusion of a sperm and egg enough to designate the resultant 'being' as human? Or, should go as far as to call it an instance of synthetic life? Should life, at it's very elemental level be broken down to just genetic transfer?

Politicians and bioethicists have grappled with these questions for more than a decade to result into the the fourteen day rule. Now that scientists have begun knocking at the door, is it the right time to revisit this rule and finally allow full scale human embryo experiments? Maybe. It all comes down to the question that we ourselves have asked numerous times in features about CRISPR technology and designer babies - Can we trust science and scientists with such technology?